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Annexure3b- Complete filing

INVENTION DISCLOSURE FORM

1. TITLE: AI-Based Soil Immune Health Monitoring and Forecasting Disease Prevention System

2. INTERNAL INVENTOR(S)/STUDENT(S): All fields in this column must be filled.

3. DESCRIPTION OF THE INVENTION: This invention relates to an Artificial Intelligence-based system aimed at persistently monitoring the biological, chemical, and physical health of agricultural soil, which is referred to as "Soil Immune Health," and predicting the onset of plant diseases from the soil before symptoms start developing in the plants. This system uses a network of affordable Internet-of-Things-based sensors in the field (these sensors monitor parameters including moisture, temperature, pH, electrical conductivity, oxidation-reduction potential, macronutrients/micronutrients, and microbial processes) along with an AI-based engine on cloud/edge computing for combining all the parameters into a real-time calculation of the dynamic Soil Immune Health Index (SIHI). (Ramson et al., 2021, p. 1)

PROBLEM ADDRESSED BY THE INVENTION: Current methods of soil evaluation – such as single-parameter sensing systems, biodegradable one-shot sensors, microbial burden kits based on membranes and laboratory analysis – are mostly reactive, offline and disjointed. The former measures just a single parameter, while the latter requires removing a physical sample for lab analysis, meaning farmers usually learn about a disease outbreak only after visual damage to crops has occurred. (Dhamu et al., 2025)

OBJECTIVE OF THE INVENTION:

For instant and non-intrusive evaluation of biological, chemical, and physical factors affecting the soil immune system.

For developing an AI/ML model combining the information from multi-parametric sensors on soil together with historical records of diseases to create a quantitative measure of Soil Immune Health Index.

To provide early alert concerning the risk of soil-borne pathogens before any visual manifestation of plant diseases.

To provide farmers with early warning along with targeted recommendations based on their specific cases using dashboards and mobile applications.

To minimize the use of pesticides and fungicides through specific intervention.

C. STATE OF THE ART/RESEARCH GAP/NOVELTY: Describe how your invention fulfills the research gap.

D. DETAILED DESCRIPTION:

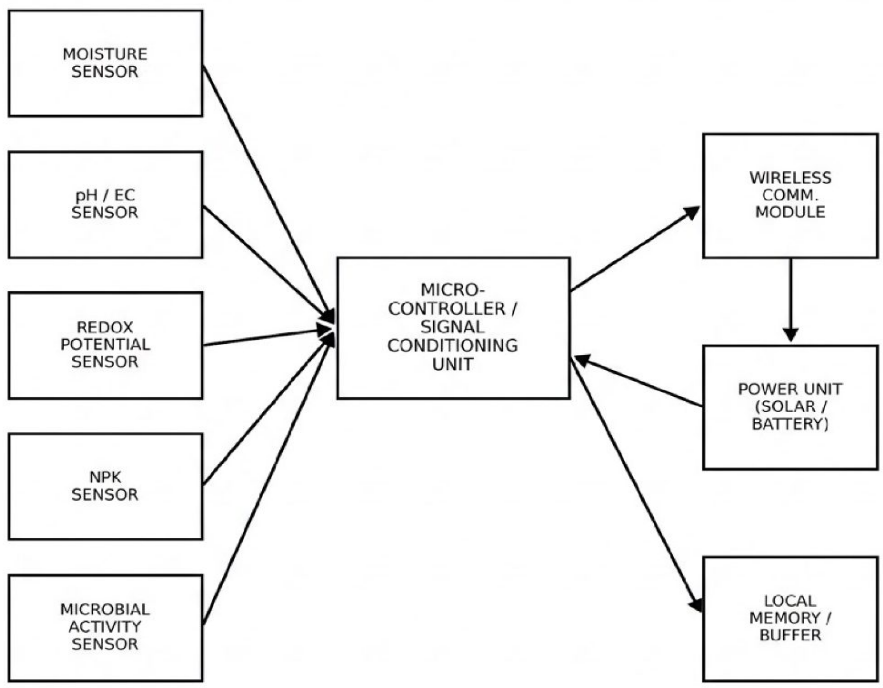
The present invention is an AI-Based Monitoring system for predicting soil health and diseases. It uses sensors, servers, as well as algorithms to monitor soil conditions and identify soil health. This current system can predict diseases based on environmental values. This method delivers farmers with knowledge that enables them to increase food production whilst minimizing the use of harmful chemicals.

The proposed invention utilizes a network of Internet of Things devices, cloud and edge computing platforms, and AI algorithms to continuously monitor selected physical, chemical, and biological properties of agricultural soil and predict the occurrence of disease before visual detection. The system is intended to provide continual assessment of the soil immune health index and related parameters, predicting potential disease states, and advising farmers on optimal interventions for sustaining soil health or managing disease outbreaks. The system can offer a comprehensive prediction model of the soil's immunological health capacity, in contrast to traditional approaches that require farmers to bring samples to a laboratory for analysis.

D.1 Multi-Parameter Soil Data Acquisition:

The first step in the process is to ensure that the soil data in the agricultural field is collected using the IoT-based sensor network. It is important to state that multiple sensors were used in order to collect the relevant information on the moisture, temperature, pH, EC, ORP, N, P, K, and micro-organisms in the soil. These sensors transmitted the data to the gateway via LoRaWAN, GSM, or Wi-Fi. At this point, the gateway receives the data and uploads it to the cloud for storage and other processing purposes. Most importantly, in this case, there were multiple points for data collection, which implies that the data was gathered in real-time. As a result, farmers can easily monitor their fields and respond to identified issues in a timely manner. For instance, with such an IoT-based system in place, it becomes possible to detect minor issues in the soil, which may imply the development of potential diseases and the lack of particular nutrients in the plants in the agricultural fields.

FIG. 2



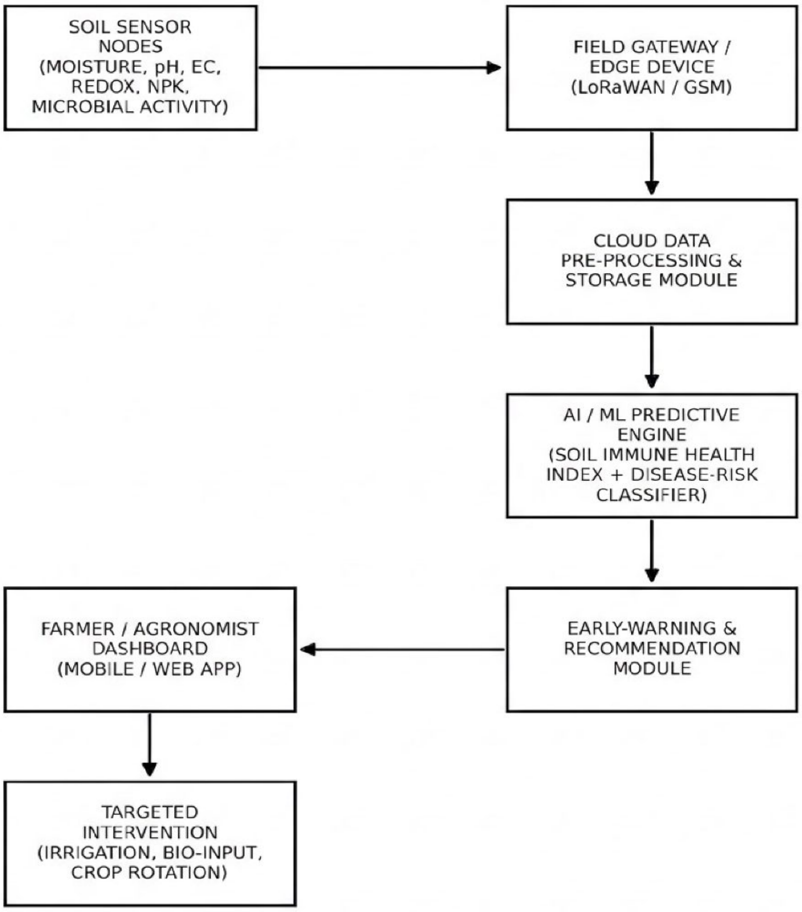
Block Diagram of a Single Soil Sensor Node

Figure 1. Multi-Parameter Soil Data Acquisition Architecture.

D.2 AI-Based Soil Immune Health Index (SIHI) Generation:

Once the information is received from the sensors, the proposed system processes and prepares the information by removing missing values, normalizing, reducing noise, and aligning information acquired from different sensors. The processed data is subjected to artificial intelligence and machine learning models to predict the accurate value of the Soil Immune Health Index (SIHI). The algorithms used include the Long Short-Term Memory network, which predicts changes in future soil states, and Random Forest and Gradient Boosting algorithms that identify the relationship between various factors and the recorded disease status. Besides using these algorithms, the system also applies ensemble learning techniques whereby different algorithms are combined to improve model performance and accuracy (Kumar et al., 2021). The resulting value of SIHI will be used to monitor and predict any early signs of disease developments.

FIG. 1



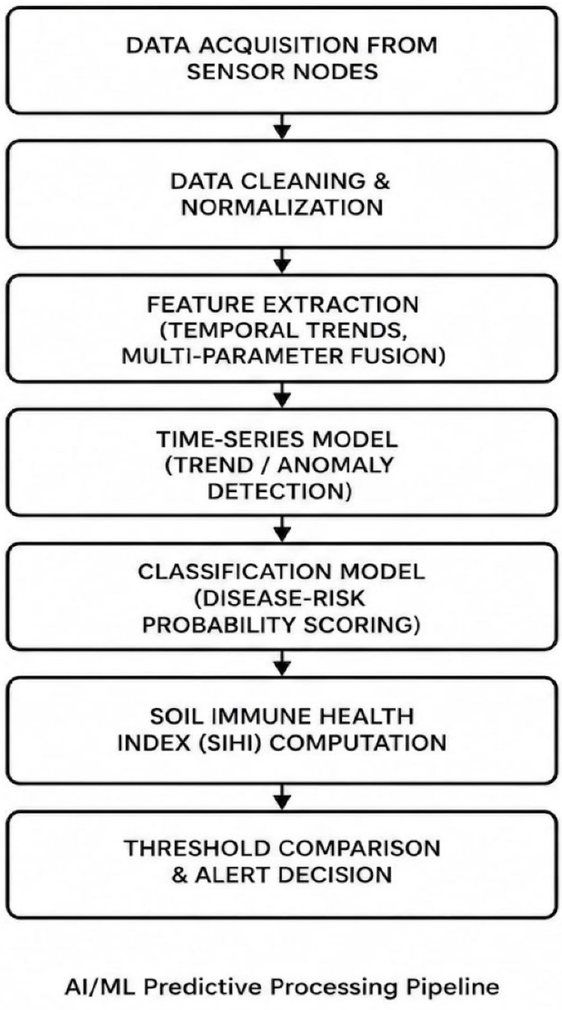
System Architecture of the AI-Based Soil Immune Health Monitoring and Predictive Disease Prevention System

Figure 2. AI-Based Soil Immune Health Index Generation Process.

D.3 Predictive Disease Detection and Risk Analysis:

By constantly updating the Soil Immune Health Index and utilizing a historical database of diseases, the proposed invention is able to predict the likelihood of diseases. The algorithm was able to take into account the changing levels of soil moisture, nutrients, pH, microorganisms, and other variables that could indicate the impending disease.

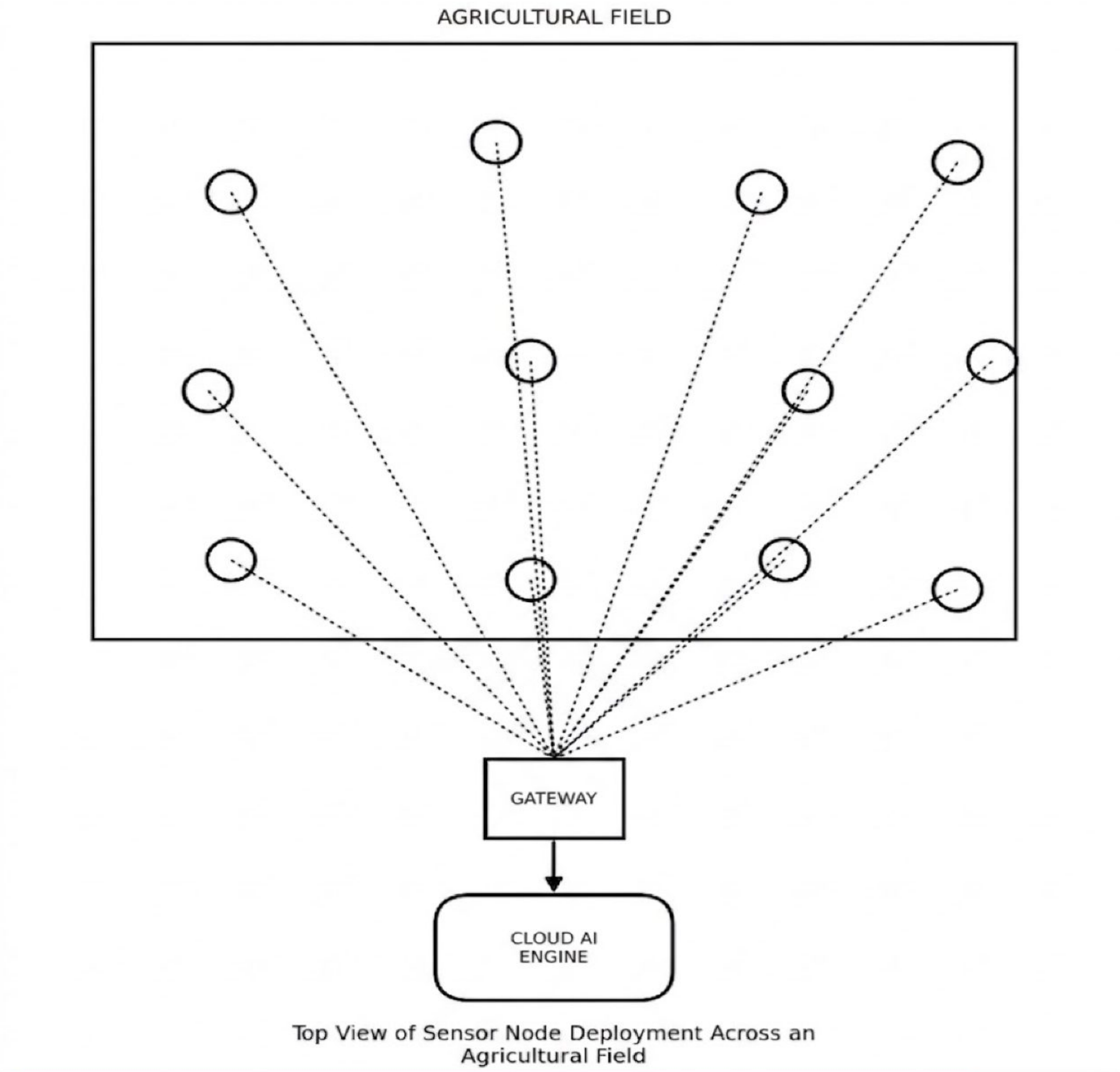
The program is able to identify Root Rot, Fusarium Wilt, Damping-off, Pythium Infection, Rhizoctonia Disease, Bacterial Wilt, Nematode Infestation, and Phytophthora Root Disease (Selvi et al., 2025). Rather than waiting for the appearance of symptoms, the algorithm is able to predict the likelihood of disease development and notify the user.



Caption: Figure 3. AI-Based Predictive Disease Detection Pipeline.

D.4 Intelligent Recommendation and Decision Support System:

Based on the calculated Soil Immune Health Index and the probability of the pathogens, the system offers a personalization recommendation that can improve the soil quality and decrease the possibility that the disease occurs. The recommendation engine suggests a course of action that has the potential to ensure that irrigation, fertilization, conditioning, rotation, and disease management practices are implemented. The system recommends that introduction of such bio-control agents as T. harzianum, P. fluorescens, B. subtilis, and bio-fertilizers, including Rhizobium, Azotobacter, Azospirillum, compost, vermicompost, biochar, neem cake, humic acid, gypsum, and agricultural lime, should be adopted to improve the soil quality and address the detected issues (Huang & He, 2023). This approach helps to increase the fertility of the land and decrease the usage of chemical agents (Rodriguez et al., 2020, pp. 13638-13650).



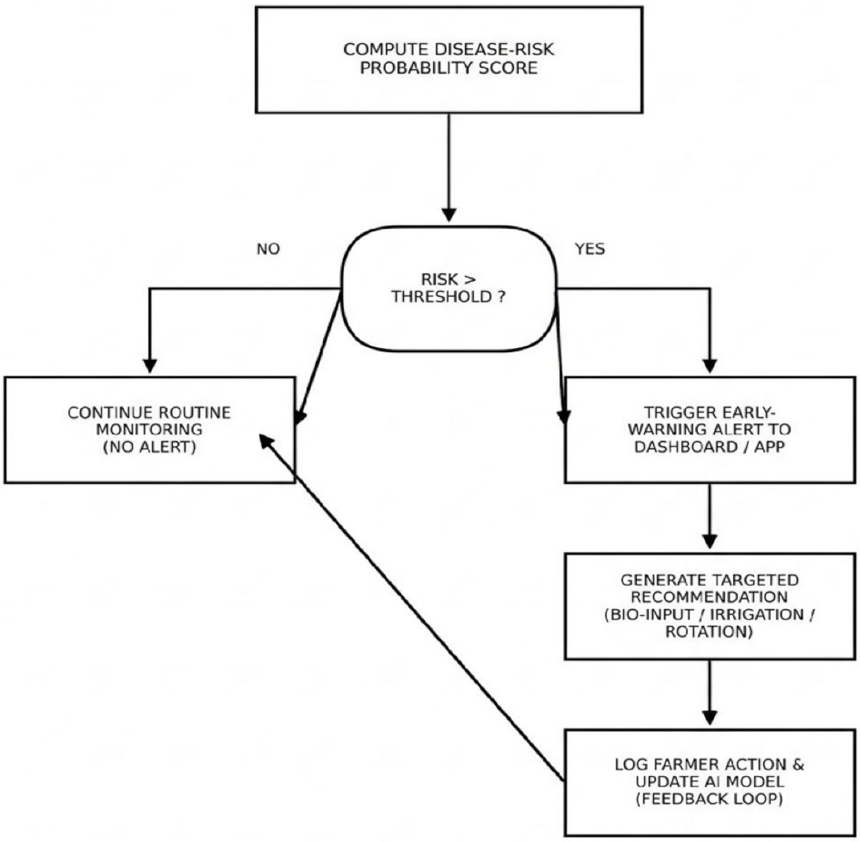
Caption: Figure 4. Intelligent Recommendation and Decision Support Workflow.

D.5 Farmer Dashboard and Alert Management:

The information is then presented on a cloud dashboard and mobile app for farmers, agronomists, and farm managers. By accessing the dashboard, one can monitor and review the Soil Immune Health Index, disease probability, and the status of all sensors. Trends in soil immunity and nutrients can also be reviewed in graphs and charts. (MapMyCrop: Crop Monitoring, 2026)

In case the probability of disease reaches a particular threshold, farmers, agronomists, and farm managers are alerted through mobile applications, SMS, and emails with all the details regarding the predicted disease, its location, probability, severity, and the necessary control measures to reduce the impact on the farm. This way, timely intervention can be done to reduce disease impact on farms. (Kaplun et al., 2024)

FIG. 5



Early-Warning Alert and Recommendation Workflow

Figure 5. Farmer Alert and Recommendation Dashboard.

D.6 Hardware Components:

“The proposed invention comprises IoT-based sensing and communication hardware such as soil moisture, pH, EC, Oxidation-Reduction Potential, NPK, microbial activity, temperature sensors, ESP32/STM32 microcontroller, LoRaWAN/GSM module, gateway, cloud server, solar panel, battery, and wireless communication for field monitoring.”

D.7 Software Components, AI Models and System Workflow:

The whole project is built with the Python programming language, TensorFlow, Scikit-learn, XGBoost libraries, the Flask or FastAPI framework, MQTT protocol, cloud database, and Android/web applications. The AI area utilizes Long Short-Term Memory Cell for time series forecasting, Random Forest, Gradient Boosting, and Ensemble Learning algorithms for classification of diseases while Autoencoders help identify anomalies. If the image capturing module is integrated into the system, then Convolutional

Neural Networks are deployed as well (Ahn et al., 2023). The system involves such steps as sensing the soil, transmitting the data to the cloud, preprocessing, calculating the PH index of the soil immune system with AI methods, predicting the disease, recommending the treatment, and informing the farmer through dashboards and mobile Applications.

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E. RESULTS AND ADVANTAGES:

Expected Results:

The benefits of the service to a grower include:

● Earlier warning of developing soil-borne disease risk, often days to weeks before visible symptoms appear on plants (Delfani et al., 2024)

● Reduction in use of fungicides or pesticides due to increasingly effective, precise applications (Zanin et al., 2022)

● Higher yields and lower input costs due to a shift from reactive to preventative treatment of disease (Papadopoulos et al., 2024)

● An up-to-date quantitative measure of the health or immunity status of the soil that can be used as a reference for future crops, allowing growers to maintain long-term soil health records (Wade et al., 2022)

Advantages:

Unlike membrane- or single-use sensors, our solution is non-invasive and permits continuous readouts. It uses multi-parameter tracking, as opposed to single-parameter readouts, and employs artificial intelligence to predict and prevent risks, as opposed to simply indicating the current state of affairs. In addition, it is easily scaled up, having been tested and proven successful on thousands of small-scale farms and successfully implemented at the commercial scale. (CropPulse — AI Crop Monitoring & Yield Prediction, n.d.) Lastly, it helps significantly reduce losses due to improper cultivation and agrochemical waste by calculating the right amount based on crop needs. (Talaviya et al., 2020)

F. EXPANSION:

● Integration with drone/satellite multispectral imagery to correlate above-ground canopy stress with below-ground SIHI trends

● Extension of the predictive model to livestock housing or aquaculture pond substrate monitoring

● Integration with automated watering/fertigation systems for closed-loop corrective action

● Blockchain-based traceability of soil health records for certification and supply-chain use cases

● A recommendation marketplace connecting predicted risk profiles with relevant bio-input suppliers.

G. WORKING PROTOTYPE/ FORMULATION/ DESIGN/COMPOSITION:

The working prototype is under development.

H. EXISTING DATA: Any clinical or comparative data necessary to support your invention. (Comparative) NA

4. USE AND DISCLOSURE (IMPORTANT): Please answer the following questions:

5. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us. NA

6. Provide the terms and conditions of the MOU, as well as whether the work is done in collaboration within or outside the university (any industry, other universities, or any other entity). NA

7. Potential Chances of Commercialization. Yes

8. List of companies that can be contacted for commercialization, along with the website link. NA

9. Any basic patent that has been used and for which we need to pay royalties to them. NA

10. FILING OPTIONS: Please indicate the level of your work that can be considered for provisional/complete/PCT filings. Complete

11. KEYWORDS:

Internet of Things (IoT) Sensors Network Edge computing Cloud computing Sensor fusion Time series prediction LSTM/DL Ensemble machine learning Real-time data analysis Pathogen detection in soil Rhizosphere monitoring Microbial activities in soil Prediction of nutrient deficiencies Monitoring of soil pH Redox potential sensing Soil electrical conductivity Prevention of root rot Prediction of wilt disease Sustainable agriculture Early warning system Decision support system Precision farming Agri-tech Farm management software Predictive maintenance of soil health Non-destructive testing Optimization of crop yield Digital agriculture platform Reduction of pesticides usage Sustainable crop protection Soil biodiversity.

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Sr. No.	Patent I'd	Abstract	Research Gap	Novelty
	US11226306B2	Discloses a microbial sen[illegible]	[illegible]	[illegible]
II.	US20210140908A1	Discloses an elongated pr[illegible]	[illegible]	[illegible]
III.	US9964532B2	Discloses a small, biodeg[illegible]	[illegible]	[illegible]
IV.	US10179926B2	Discloses membrane-bas[illegible]	[illegible]	[illegible]
V.	US12029147B2	The invention discloses a[illegible]	[illegible]	[illegible]
VI.	US20220012383A1	Discloses a precision-agri[illegible]	[illegible]	[illegible]

Have you described or shown your invention design to anyone or in any conference?	YES ( )	NO (✓)
Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturi	YES ( )	NO (✓)
Has your invention been described in any printed publication, or any other form of media, such as the Internet?	YES ( )	NO (✓)
Do you have any collaboration with any other institute or organization on the same? Provide name and other details.	YES ( )	NO (✓)
Name of Regulatory body or any other approvals if required.	YES ( )	NO (✓)